

Java Binary Assignments

Part (a): Integer to Binary

```
import java.util.Scanner;

public class BinaryConverter {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter an integer: ");
        int num = input.nextInt();

        String binary = "";
        while (num > 0) {
            binary = (num % 2) + binary;
            num = num / 2;
        }
        System.out.println("Binary: " + binary);
    }
}
```

Part (b): Array Function

```
public static int[] convertToBinaryArray(int n) {
    int temp = n, count = 0;
    while (temp > 0) { temp /= 2; count++; }
    int[] binaryArray = new int[count];
    for (int i = count - 1; i >= 0; i--) {
        binaryArray[i] = n % 2;
        n /= 2;
    }
    return binaryArray;
}
```

Part (c): String Function

```
public static String getBinaryString(int n) {  
    StringBuilder binary = new StringBuilder();  
    while (n > 0) {  
        binary.insert(0, n % 2);  
        n /= 2;  
    }  
    return binary.toString();  
}
```